Some strange spinner...

X23 (2023) — English translation

Consider the mechanical system shown in the figure. Two small spheres A and B of mass m are fixed at the ends of a homogeneous rod of length $2l$ and mass $4m$, attached at the middle perpendicular to the horizontal axis. Above the initial position of the sphere A, in which the rod is horizontal and motionless, there is a vertical tunnel from which at specially selected moments of time a ball of mass m flies out, colliding with the spheres and causing the system to rotate around its axis. The speed of the ejected ball just before the collision is v. As a result of the first collision, the system acquires a certain angular velocity ω_1. When the sphere B reaches the position in which the sphere A was originally located, B collides with a new ball, and the system acquires some angular velocity ω_2, and so on. In all parts of the problem, consider all collisions to be elastic. Neglect the sizes of balls and spheres.

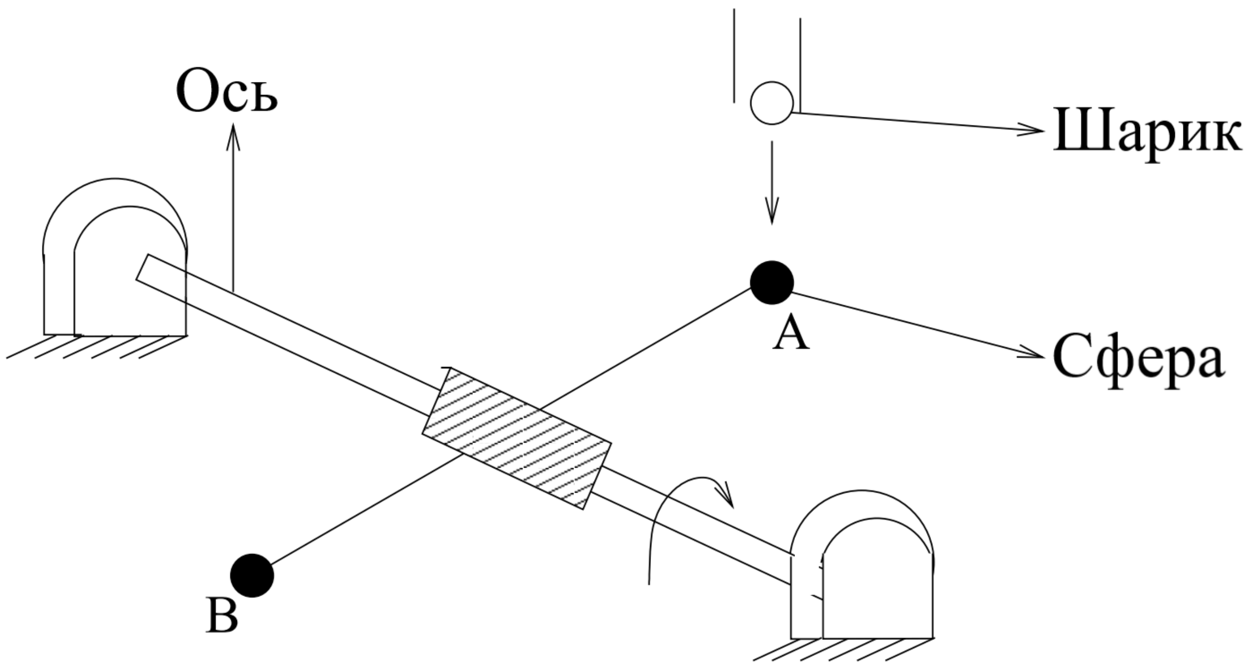

Fig. 1.

Part A. Central collisions without dissipation (2.0 points)

In this part of the problem, all collisions are central. There are no dissipations in the system.

A1	Determine the moment of inertia of the system of the rod and spheres relative to the axis of rotation. Express your answer in terms of m and l.	**0.30**

A2	Express the angular velocity ω_{i+1} of the system after the collision with the $i+1$ emitted ball in terms of ω_i, v and l.	**0.50**

A3	What is ω_1 equal to? Express your answer in terms of v and l.	**0.20**

As can be expected, the angular velocity of such a rotating system tends at $i \to \infty$ to a constant value ω^*.

A4	Express ω^* in terms of v and l.	**0.20**

A5	By solving the recurrence equation obtained in step **A1**, you will obtain an explicit expression for ω_i. Express your answer using i, v and l. Hint: Use the variable $\omega_i' = \omega_i - v/l$.	**0.60**

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Let now, instead of one pair of spheres, we have two pairs of them, placed on rods perpendicular to each other and the axis.

A6	What would be the limiting value ω^* for such a system?	**0.20**

Part B. Off-center collisions without dissipation (3.0 points)

Let's go back to the system with one pair of spheres. In this part of the problem, non-central collisions of balls with spheres A and B are studied. There are no dissipations in the system (including friction between the escaping ball and the spheres). We will characterize the off-center impact by the parameter $k = b/r$, where $b < r$ is the impact parameter with which the ejected ball hits the sphere, and $r \ll l$ is the distance between the centers of the ball and the sphere when they come into contact (see Fig.).

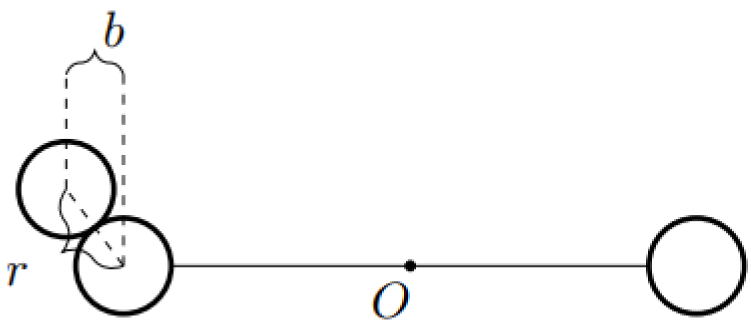

Fig. 2.

B1	At what value of the parameter k_0 will the speed of the first emitted ball, immediately after colliding with the sphere, be directed horizontally?	**1.00**

Next, an arbitrary value of the parameter k is considered.

B2	Express the angular velocity ω_{i+1} of the system after the collision with the $i+1$ emitted ball in terms of ω_i, v, l and k.	**1.00**

B3	What is ω_1 equal to? Express your answer in terms of v_0, l and k.	**0.20**

B4	By solving the recurrence equation obtained in step **B2**, you will obtain an explicit expression for ω_i. Express your answer using i, v and l.	**0.80**

Part C. Based on Kotkin–Serbo (4.0 points)

In this part of the problem, the motion of the system under the influence of friction is studied. Under the influence of friction, a moment M acts on the system, which is a certain function of the angular velocity ω. Depending on the geometry of the system, the main contribution comes from: Dry friction – moment $M = \alpha I = \text{const}$; Linear viscous friction – moment $M = \beta I\omega$, $\beta = \text{const}$; Quadratic viscous friction – moment $M = \gamma I\omega^2$, $\gamma = \text{const}$. Here I is the moment of inertia of the system. Instead of the moment of inertia of the system and the speed of the ball, we introduce parameters $\varepsilon = \frac{ml^2}{I}$ and $\omega_0 = \frac{v}{l}$. At all points in part C the impacts are central and elastic. Express all answers in points C1 – C3 using ε and ω_0.

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Express all answers in points C1 – C3 through ε and ω_0.

C1	Find at what values of the parameter α the system in the dry friction mode will continue to move indefinitely long after the first collision with the ball.	**0.50**

C2	Find at what values of the parameter β the system in the mode of linear viscous friction will continue to move indefinitely long after the first collision with the ball.	**0.50**

C3	Find at what values of the parameter γ the system in the mode of quadratic viscous friction will continue to move indefinitely long after the first collision with the ball.	**0.50**

Let us now proceed directly to the exact calculation of the steady-state average angular velocities in each of the three modes. To do this, together with the equation for a collision with a ball, you need to solve the equation of free motion of the system and find stationary points.

C4	Find the steady-state average over the period angular velocity of the system $\overline{\omega}_1$ in the dry friction mode. Express your answer in terms of ε, α and ω_0.	**0.70**

Express your answers in C5 and C6 using ε, β and ω_0.

C5	Find the stationary value of the angular velocity $\omega_2^{\uparrow}$ of the system immediately after the next collision with the ball in the mode of linear viscous friction.	**0.50**

C6	Find the steady-state average over the period angular velocity of the system $\overline{\omega}_2$ in the mode of linear viscous friction.	**0.40**

Express your answers in C7 and C8 in terms of ε, γ and ω_0.

C7	Find the stationary value of the angular velocity $\omega_3^{\uparrow}$ of the system immediately after the next collision with the ball in the quadratic viscous friction mode.	**0.50**

C8	Find the steady-state average over the period angular velocity of the system $\overline{\omega}_3$ in the mode of quadratic viscous friction.	**0.40**

Source: https://pho.rs/p/3160